Cocoercivity does not give a quadratic residual rate for Douglas–Rachford splitting

Exact counterexamples, the sharp worst-case exponent, and limits of scalar relaxation

Anonymous

5 November 2026

Status. Unrefereed, AI-assisted research note. Internal AI-assisted checking is not external peer review or unaffiliated reproduction. The contribution and literature boundary are stated in Section 7. The verification record distinguishes exact finite certificates, written universal arguments and the external theorems they use.

Version: 0.1.0-candidate. Archived evidence and DOI. Source and verification package.

Abstract

Conjecture 3.5 of Abbaszadehpeivasti and Zamani (2025, version 2) proposes an order N^{-2} bound for the squared fixed-point residual of Douglas–Rachford splitting when one operator is a convex subdifferential and the other is cocoercive. We give two exact refutations. A scalar quadratic example defeats part of the stated overrelaxation range at the second iterate. A two-dimensional cocoercive rotation defeats the claim at unit relaxation, where the substantive issue is the absence of gradient structure. For every fixed admissible stepsize and relaxation parameter, the worst-case squared residual over the stated class has order N^{-1}. For the $A = 0$ subclass, allowing arbitrary real scalar multipliers and measuring the final unrelaxed residual, we derive an exact finite-horizon minimax value including adaptive and random choices. We also describe prescribed schedules over linear operators and obtain an elementary exact infimum for that linear class with multipliers restricted to $(0, 2)$. These latter results combine a classical contraction polynomial inequality, the known sharp firmly nonexpansive residual bound, and a rotation-product construction already present in recent literature. A rational hinge example shows that the full mixed class can exceed the maximum of a scalar proximal example and the pure linear-rotation envelope. The paper's separate objective-value acceleration conjectures remain unresolved.

1. Problem and contribution

Douglas–Rachford splitting seeks a solution of $0 \in Ax + Bx$ by evaluating resolvents of the two operators separately. This can make a coupled problem easier to handle. A central theoretical question is how many updates are needed before the iteration changes very little. Here we test one proposed uniform guarantee for that stopping quantity, not the practical speed of all splitting solvers.

All spaces in the stated theorems are finite-dimensional real Euclidean spaces. The dimension may vary in a worst-case supremum; every lower construction below needs at most two dimensions. Norms are Euclidean, and matrix norms are their induced operator norms. Write $J_C = (I + C)^{-1}$

for the resolvent of a maximally monotone operator C, and $R_C = 2J_C - I$ for its reflected resolvent.

Given maximally monotone A and B, Douglas–Rachford splitting uses

$$x^k = J_{\gamma B} w^k, \qquad y^k = J_{\gamma A}(2x^k - w^k), \qquad w^{k+1} = w^k + \lambda(y^k - x^k). \tag{1}$$

Here $\gamma > 0$ and $0 < \lambda < 2$. Let T_λ denote its fixed-point map. The residual at index N is

$$r_N = \|T_\lambda w^N - w^N\| = \|w^{N+1} - w^N\|.$$

There are $N - 1$ completed updates before evaluating this residual. Keeping this convention matters when comparing exact constants.

The residual measures an approximate fixed-point condition for the splitting variable w. It is not an objective gap or the distance of the primal shadow $x = J_{\gamma B} w$ to a solution. Converting it into those quantities requires further assumptions, such as an appropriate error bound. The initial-distance normalization below is an analytical scale, not an implementable stopping test, and update counts do not account for the cost of evaluating resolvents.

An everywhere-defined operator B is β-cocoercive when

$$\langle Bx - By, x - y\rangle \geq \beta\|Bx - By\|^2 \quad \text{for every } x, y. \tag{2}$$

This property permits a rotational component; it does not require B to be a convex gradient. For a closed proper convex function g, its subdifferential is denoted ∂g.

The target is Conjecture 3.5 in the inspected latest arXiv version of Abbaszadehpeivasti and Zamani (2025). Its additional hypotheses are $A = \partial g$, B is β-cocoercive, $0 < \gamma < \beta$, and T_λ has a fixed point $w^\star$. It asserts

$$r_N^2 \leq \frac{\lambda^2}{((N-1)\lambda + 1)^2}\|w^1 - w^\star\|^2. \tag{C}$$

This note establishes that (C) is false and identifies the correct uniform exponent under these assumptions. It does not determine the exact worst-case constant for the entire nonlinear mixed class. The exact minimax result in Section 5 concerns an explicitly restricted algorithm class, with $A = 0$.

Result	Operator and algorithm class	Error measure and conclusion
Proposition 1	Scalar convex quadratics; overrelaxation	Exact failure of (C) at $N = 2$
Proposition 2	$A = 0$; cocoercive planar rotation; $\lambda = 1$	Exact failure of (C); the fixed instance still converges geometrically
Theorem 4	Full mixed class in (C); fixed admissible parameters	Squared update residual has uniform worst-case order N^{-1}, hence residual norm order $N^{-1/2}$; no sharp full-class constant
Theorem 5	Linear B, $A = 0$; prescribed real scalars	Exact final unrelaxed-residual envelope after m updates

Result	Operator and algorithm class	Error measure and conclusion
Theorem 6	Possibly nonlinear B, $A = 0$; adaptive real scalars	Exact unrelaxed-residual minimax after m updates; not generic mixed DRS tuning
Corollary 6.1	Linear B, $A = 0$; adaptive scalars in $(0, 2)$; $c > 2$	Exact minimax infimum, not attained in the open interval
Section 6 and Appendix A	Mixed splitting with a hinge and rotation	Exact finite witnesses against an elementary repair considered here

In the scheduling results $m = N - 1$, but the reported error is $\|G(w^m)\|$, without multiplying by a reporting relaxation. This convention prevents a small scalar from making an unchanged state appear accurate.

2. Two exact counterexamples

Proposition 1: a scalar overrelaxation obstruction

Set $\beta = 1$, $\gamma = 1/2$, $\lambda = 9/5$, $B = 0$ and $g(x) = 5x^2$. Thus $A = 10I$, and both operators are subdifferentials of smooth convex functions. The resolvents are $J_{\gamma B} = I$ and $J_{\gamma A} = I/6$. Consequently,

$$T_\lambda = (1 - \lambda)I + \frac{\lambda}{6}I = -\frac{1}{2}I.$$

With $w^1 = 1$ and $w^\star = 0$, the second residual is

$$|T_\lambda w^2 - w^2|^2 = \frac{9}{16} > \frac{81}{196} = \frac{\lambda^2}{(\lambda + 1)^2}. \tag{3}$$

The ratio is $49/36$. Every parameter lies strictly inside the proposed intervals. Since the zero operator is β-cocoercive for every positive β, no limiting argument is needed. This refutation concerns the stated relaxation range. It does not explain what happens at $\lambda = 1$.

Proposition 2: the unit-relaxation obstruction

Let $\beta = 1$, $\gamma = 1/2$, $\lambda = 1$, $g \equiv 0$, and

$$B = \frac{1}{50}\begin{pmatrix} 1 & -7 \\ 7 & 1 \end{pmatrix}, \qquad w^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad w^\star = 0. \tag{4}$$

Then $A = \partial g = 0$. Direct multiplication gives

$$B^\mathsf{T} B = \frac{B + B^\mathsf{T}}{2} = \frac{1}{50}I.$$

Therefore $\langle Bh, h\rangle = \|Bh\|^2$ for every h: B is exactly 1-cocoercive. It is continuous, everywhere defined and monotone, hence maximally monotone. It is invertible, so the solution and fixed point are uniquely zero.

Since $A = 0$ and $\lambda = 1$, the iteration reduces to $T = J_{B/2}$. Its matrix is

$$T = \frac{2}{205}\begin{pmatrix} 101 & 7 \\ -7 & 101 \end{pmatrix}, \quad T^{\mathsf{T}}T = \frac{40}{41}I, \quad (I-T)^{\mathsf{T}}(I-T) = \frac{1}{205}I. \tag{5}$$

It follows that, for every positive integer N,

$$r_N^2 = \frac{1}{205}\left(\frac{40}{41}\right)^{N-1} = \frac{1}{200}\left(\frac{40}{41}\right)^{N}. \tag{6}$$

At $N = 20$, (C) would give $r_{20}^2 \le 1/400$. Instead,

$$\left(\frac{40}{41}\right)^{20} = \left(1 - \frac{1}{41}\right)^{20} \ge 1 - \frac{20}{41} = \frac{21}{41} > \frac{1}{2}.$$

Thus (6) is strictly larger than $1/400$. Its numerical value is approximately 0.00305135471429, compared with 0.0025, an excess of about 22.05%. The numerical value is illustrative; the rational identities and Bernoulli inequality prove the claim. For this fixed matrix, the first violating index is 18, as checked separately by exact arithmetic.

The matrix is a positive dilation plus a skew-symmetric rotation. It is not a convex gradient: its Jacobian is not symmetric. Cocoercivity controls the rotation without eliminating it. This distinguishes Proposition 2 from the scalar relaxation issue in Proposition 1. It does not prove that adding a gradient assumption repairs (C) at unit relaxation; that possible residual bound is not established here.

A necessary distinction. This fixed example converges geometrically. It is not an example of a single fixed problem whose asymptotic residual decays exactly as $N^{-1/2}$. The uniform lower bound proved next selects an instance for each horizon.

3. The correct worst-case exponent

Lemma 3: cocoercivity of the resolvent residual

Let B be β-cocoercive and $\gamma > 0$. Define

$$J = J_{\gamma B}, \qquad G = I - J, \qquad c = 1 + \frac{\beta}{\gamma} > 1. \tag{7}$$

Then G is c-cocoercive, and $R = I - 2cG$ is nonexpansive.

Proof. For two inputs write $z = Jv - Jw$ and $d = Gv - Gw$. The resolvent identity gives $d = \gamma(BJv - BJw)$ and $v - w = z + d$. Hence

$$\langle d, v - w\rangle = \langle d, z\rangle + \|d\|^2 \ge \left(\frac{\beta}{\gamma} + 1\right)\|d\|^2.$$

Expanding $\|(v-w) - 2cd\|^2$ proves nonexpansiveness. This is the standard resolvent averagedness relationship; see Giselsson (2017, Proposition 5.2), after translating the reciprocal cocoercivity convention. We include the proof to fix the constants. □

Theorem 4: order N^{-1} is sharp under the conjecture's assumptions

Fix $\beta > 0$, $0 < \gamma < \beta$ and $0 < \lambda < 2$, and set $c = 1 + \beta/\gamma$. Let W_N be the supremum of r_N^2 over the class in Section 1, with $\|w^1 - w^\star\| \le 1$. For

$$N \ge \frac{c^2}{\lambda(2c-\lambda)}, \qquad N \ge 2,$$

one has

$$\frac{\lambda}{2c-\lambda}\frac{(N-1)^{N-1}}{N^N} \le W_N \le \frac{\lambda}{(2-\lambda)N}. \tag{8}$$

In particular, $W_N = \Theta(N^{-1})$ for every fixed admissible triple of parameters. The proposed order N^{-2} cannot hold with any uniform constant on this class.

Upper bound. Put $S = R_{\gamma A}R_{\gamma B}$, a nonexpansive map. Then $T_\lambda = (1-\lambda/2)I + (\lambda/2)S$. The averaged-map identity gives

$$\|T_\lambda w - w^\star\|^2 \le \|w - w^\star\|^2 - \frac{2-\lambda}{\lambda}\|T_\lambda w - w\|^2. \tag{9}$$

Summing (9) for $k = 1, \dots, N$ bounds the sum of squared residuals by $\lambda/(2-\lambda)$. Nonexpansiveness of T_λ makes the residual norms nonincreasing. Thus Nr_N^2 is at most this sum. This is a standard Fejér-monotonicity argument, not a new general upper bound.

Lower bound. Take $A = 0$ and identify the real plane with the complex numbers. Let G multiply by a complex number z satisfying

$$\operatorname{Re} z = c|z|^2, \qquad q = |z|^2 \in [0, c^{-2}].$$

Then $B = \gamma^{-1}G(I-G)^{-1}$ is a real linear β-cocoercive operator. To check this, $I - G$ is invertible because $|z| \le 1/c < 1$, and its multiplier $b = z/[\gamma(1-z)]$ satisfies

$$\operatorname{Re} b = \frac{(c-1)q}{\gamma|1-z|^2} = \beta|b|^2. \tag{10}$$

The relaxed map multiplies by $1 - \lambda z$. Starting at a unit vector,

$$r_N^2 = \lambda^2 q\,[1 - \lambda(2c-\lambda)q]^{N-1}. \tag{11}$$

Choose $q = 1/[\lambda(2c-\lambda)N]$, which lies in the allowed interval under the theorem's condition. Substitution proves the lower bound. Finally,

$$N\frac{(N-1)^{N-1}}{N^N} = \left(1 - \frac{1}{N}\right)^{N-1} \longrightarrow e^{-1},$$

so both sides of (8) have the claimed order. □

The horizon-dependent lower examples can each be strongly monotone and geometrically convergent, while their strong-monotonicity moduli deteriorate as the horizon grows. There is no contradiction with linear-rate theorems that impose a uniform positive modulus.

The exact leading constant for the full mixed class remains unknown here. In particular, the lower constant in (8) is not asserted to be a universal sharp constant.

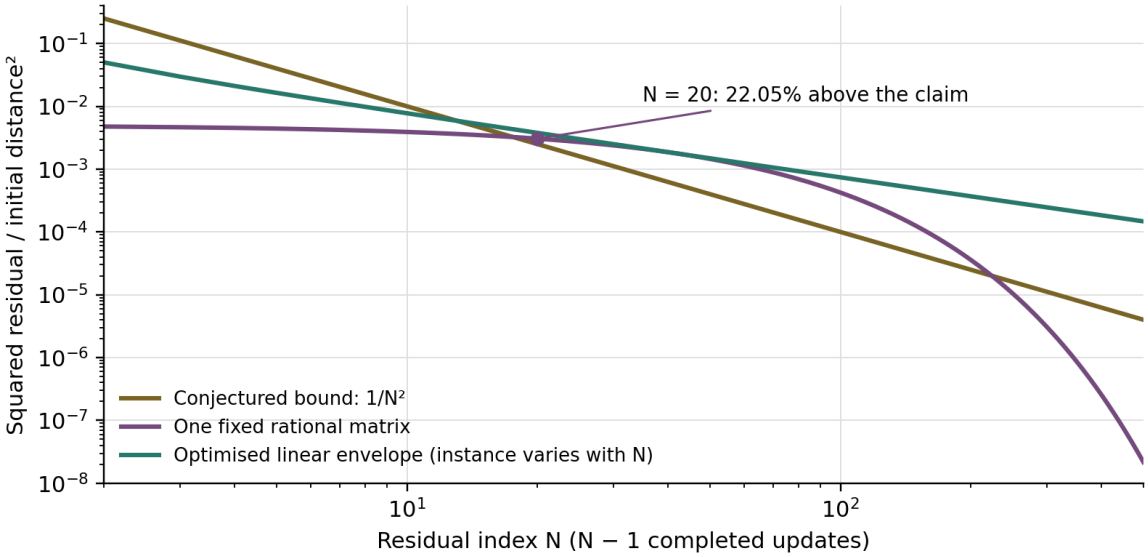


Figure 1: At $\beta = 1$, $\gamma = 1/2$, $\lambda = 1$, the fixed rational witness crosses the proposed bound at finite horizons and eventually decays geometrically. The optimised linear envelope uses a different instance at each horizon and has order N^{-1}. The curves illustrate exact formulas; the proofs do not rely on the plot.

4. Exact performance of prescribed schedules on linear operators

The preceding construction also gives a complete linear-class calculation. This section is an explicit corollary of contraction polynomial theory and known rotation constructions. It is useful for identifying exactly what scalar scheduling can change.

Fix $A = 0$, $\beta, \gamma > 0$, and use the notation (7). Consider m updates

$$w^{j+1} = w^j - \lambda_j G(w^j), \qquad j = 0, \dots, m-1.$$

Here the final error measure is the **unrelaxed** residual $\|G(w^m)\|$. This avoids the artificial possibility of shrinking a reported update by setting its reporting relaxation to zero.

Theorem 5: every prescribed real scalar schedule

For any prescribed real numbers $\lambda_0, \dots, \lambda_{m-1}$, the exact supremum of

$$\frac{\|G(w^m)\|^2}{\|w^0 - w^\star\|^2}$$

over all finite-dimensional linear β-cocoercive B and initial points with $w^0 \neq w^\star$ is

$$\max_{0 \le q \le c^{-2}} q \prod_{j=0}^{m-1} [1 - (2c\lambda_j - \lambda_j^2)q]. \tag{12}$$

Each value in the maximisation is attained by a real two-dimensional operator of the form (10), with scalar endpoints included.

Proof. By Lemma 3, $R = I - 2cG$ is a linear contraction. The residual after m steps is the polynomial operator

$$p(G) = G \prod_{j=0}^{m-1} (I - \lambda_j G).$$

Von Neumann's polynomial inequality bounds its norm by the maximum of $|p(z)|$ over the disk with centre $1/(2c)$ and radius $1/(2c)$. An inspected modern statement and proof are in Hartz (2025, Proposition 2.3), specialised to scalar coefficients. For real matrices, apply the inequality to their complexification, which preserves the Euclidean operator norm. The maximum-modulus principle puts the maximum on the boundary circle $\operatorname{Re} z = c|z|^2$. On that circle,

$$|1 - \lambda_j z|^2 = 1 - (2c\lambda_j - \lambda_j^2)|z|^2.$$

This proves the upper bound (12). Multiplication by a boundary z on the real plane attains the scalar expression, and (10) supplies an admissible B. □

For constant $0 < \lambda < 2c$, put $a = \lambda/(2c)$ and $N = m + 1$. Multiplying (12) by λ^2 gives the exact squared update-residual envelope

$$E_N(a) = \begin{cases} \dfrac{a}{1-a}\dfrac{(N-1)^{N-1}}{N^N}, & N \geq [4a(1-a)]^{-1}, \\ 4a^2|1-2a|^{2N-2}, & N < [4a(1-a)]^{-1}. \end{cases} \tag{13}$$

For $N = 1$, the value is $4a^2$, with the continuous $0^0 = 1$ convention when needed. Equation (13) follows by maximising the single-variable function in (11); it is not an upper bound for arbitrary nonlinear mixed splittings.

At $\beta = 1$, $\gamma = 1/2$, $\lambda = 1$, one has $a = 1/6$ and

$$E_N(1/6) = \frac{1}{5}\frac{(N-1)^{N-1}}{N^N} \quad (N \geq 2).$$

5. Exact minimax over adaptive scalar policies

The word *policy* below concerns only the scalar multiplying the current residual. It may use the entire history, complete problem information or internal randomisation. Random policies are assumed measurable, so the nonnegative residual has a well-defined, possibly infinite expectation. The iterate must still obey the displayed update, and the returned point must be the last iterate. Anchoring, a matrix update, an additional search direction and an averaged or extrapolated output are outside this class.

In the original regime $\gamma < \beta$, the scalar $c = 1 + \beta/\gamma$ exceeds two. The next theorem permits that scalar: it concerns $A = 0$ and unrestricted real multipliers, not generic mixed DRS with relaxation in $(0, 2)$. Its last-iterate error is the unrelaxed residual. Corollary 6.1 treats the conventional interval only for linear operators.

Theorem 6: a sharp adaptive limit and an exact penalty

Fix an integer $m \geq 1$, $\beta, \gamma > 0$, and c from (7). Over all policies of the form

$$w^{j+1} = w^j - \lambda_j G(w^j), \qquad \lambda_j \in \mathbb{R},$$

the minimax expected squared final residual over all everywhere-defined β-cocoercive B with a zero and initial distance at most one is exactly

$$\mathcal{M}_m = \frac{1}{c^2}\frac{m^m}{(m+1)^{m+1}}. \tag{14}$$

The constant policy $\lambda_j = c$ guarantees this bound uniformly over all such operators, including nonlinear operators, and attains the minimax value. A single two-dimensional instance, chosen for the horizon but independently of the policy, gives the following pathwise identity for $w^0 \neq 0$:

$$\frac{\|G(w^m)\|^2}{\|w^0\|^2} = \mathcal{M}_m \prod_{j=0}^{m-1}\left[1 + \frac{(\lambda_j/c - 1)^2}{m}\right]. \tag{15}$$

Lower bound and penalty. Choose the complex multiplier

$$z_\star = \frac{1 + i\sqrt{m}}{c(m+1)}.$$

It satisfies $\operatorname{Re} z_\star = c|z_\star|^2$ and corresponds to an admissible B through (10). For every real λ,

$$|1 - \lambda z_\star|^2 = \frac{m}{m+1}\left[1 + \frac{(\lambda/c - 1)^2}{m}\right].$$

Multiply this identity along the realised trajectory and use $|z_\star|^2 = 1/[c^2(m+1)]$. This proves (15), even when the scalars depend on the entire preceding trajectory. Every factor is at least one. Taking expectations proves the lower bound for random policies as well.

Upper bound. Lemma 3 makes $T = I - cG$ firmly nonexpansive. Every everywhere-defined firmly nonexpansive map T is the resolvent of the maximally monotone relation $T^{-1} - I$. Thus the sharp proximal-point bound of Gu and Yang (2019), equation (2.6), Proposition 4.1 and equation (4.4), applies. Their index N is our m: the measured difference is between iterates $m+1$ and m, after m completed updates from the initial point. In our notation the inherited bound is

$$\|(I - T)T^m w^0\|^2 \leq \frac{m^m}{(m+1)^{m+1}}\|w^0 - w^\star\|^2. \tag{16}$$

Its use here requires no linearity. Since $I - T = cG$, (16) proves the matching upper bound. □

Equality in (15) requires $\lambda_j = c$ at every step on this adversarial instance. This does not make the constant policy the unique optimal policy on all other instances. A policy may act differently on easier instances and retain the same worst-case value.

For the original parameter regime $\gamma < \beta$, one has $c > 2$. The attaining policy therefore lies outside the generic Douglas–Rachford interval $0 < \lambda < 2$. It is valid for the $A = 0$ cocoercive subclass, not as a proposed universal choice for nonzero A. Even allowing this enlarged scalar-policy class does not improve the order beyond m^{-1} for squared residual. This gives a fortiori a lower barrier for the smaller policy class.

Theorem 6 is a synthesis with an exact refinement. Sarkar and Aggarwal (2026, Appendix A) already derive the corresponding rotation product for prescribed Krasnoselskii–Mann schedules; Gu and Yang supply (16). The exact adaptive quantifiers and the penalty identity were not located together in the inspected literature. That bounded finding does not establish historical priority.

Corollary 6.1: conventional-range scalar policies on the linear subclass

Let $c > 2$ and $m \geq 1$. Restrict to finite-dimensional linear β-cocoercive B, set $A = 0$, and normalize the initial distance to a zero by one. Allow adaptive or randomized policies as in Theorem 6, but require every realized scalar to lie in the open interval $(0, 2)$. The infimum, over these policies, of the worst-case expected squared final **unrelaxed** residual is

$$L_m = \max_{0 \leq q \leq c^{-2}} q[1 - 4(c-1)q]^m = \begin{cases} \dfrac{m^m}{4(c-1)(m+1)^{m+1}}, & m+1 \geq \dfrac{c^2}{4(c-1)}, \\ c^{-2}\left(\dfrac{(c-2)^2}{c^2}\right)^m, & m+1 < \dfrac{c^2}{4(c-1)}. \end{cases} \tag{16a}$$

No admissible policy attains this infimum. Constant scalars tending to two from below approach it.

Proof. Put $d = 4(c-1)$. Since $c > 2$, we have $dc^{-2} < 1$. Let $q_* > 0$ maximize the displayed continuous function and choose once a planar boundary multiplier z with $\operatorname{Re} z = c|z|^2$ and $|z|^2 = q_*$. The construction (10) gives an admissible linear B with zero at the origin. Choose a unit initial point. For every realized $\lambda \in (0, 2)$, the function $2c\lambda - \lambda^2$ is strictly increasing on this interval and strictly smaller than d. Hence, pathwise,

$$\|G(w^m)\|^2 = q_* \prod_{j=0}^{m-1} [1 - (2c\lambda_j - \lambda_j^2)q_*] > q_*(1 - dq_*)^m = L_m.$$

The same instance is used for every policy at this horizon. All factors on the right are positive, so the strict inequality holds for every realization, even when the scalars depend on full problem information. Its expectation is strictly larger than L_m (or infinite). Thus every admissible policy has worst-case expected residual strictly larger than L_m, proving both the lower bound on the infimum and nonattainment.

Conversely take deterministic constant policies $\lambda_j = \ell$ with $0 < \ell < 2$. Theorem 5 gives their exact linear-class worst-case value as $\max_{0 \leq q \leq c^{-2}} q[1 - (2c\ell - \ell^2)q]^m$. As $\ell \uparrow 2$ the functions converge uniformly on this compact interval to $q(1 - dq)^m$. Their maxima therefore converge to L_m, proving the matching upper bound on the infimum.

Finally, $\frac{d}{dq}\{q(1 - dq)^m\} = (1 - dq)^{m-1}[1 - d(m+1)q]$. The maximizer is $q_* = 1/[d(m+1)]$ if this lies in $[0, c^{-2}]$, and otherwise is $q_* = c^{-2}$. Substitution gives (16a); the two formulas agree at the threshold. □

This is an elementary consequence of the existing rotation product and linear polynomial bound, not a separate major novelty claim. It supplies a lower barrier for any larger operator class containing these linear examples. It does **not** provide a matching upper bound for nonlinear B or for nonzero A, and the excluded boundary scalar $\lambda = 2$ is not an admissible attaining policy.

6. A hinge example beyond the two elementary envelopes

The lower family in Theorem 4 does not exhaust the finite-horizon behaviour of the mixed class. At $\beta = 1$, $\gamma = 1/2$, $\lambda = 1$, take

$$B = \frac{1}{65}\begin{pmatrix} 1 & -8 \\ 8 & 1 \end{pmatrix}, \quad g(x) = \frac{10}{59}\max\{x_1, 0\}, \quad w^1 = \begin{pmatrix} 1 \\ 2/25 \end{pmatrix}. \tag{17}$$

Again $B^{\mathsf{T}}B = (B+B^{\mathsf{T}})/2 = I/65$, so B is 1-cocoercive. The origin is the unique solution because B is strongly monotone and $0 \in \partial g(0)$. The proximal map of a scalar hinge $h(t) = a\max\{t, 0\}$ is exactly

$$\operatorname{prox}_{\gamma h}(v) = \begin{cases} v, & v < 0, \\ 0, & 0 \le v \le \gamma a, \\ v - \gamma a, & v > \gamma a. \end{cases} \tag{18}$$

Apply (18) to the first coordinate and leave the second coordinate unchanged. Ten direct rational iterations give

$$\frac{r_{10}^2}{\|w^1\|^2} = 0.0100992483946696\ldots > \frac{1}{100}. \tag{19}$$

The decimal is only a readable representation. The accompanying certificate computes the rational numerator and denominator from (1), tests every branch of (18), and verifies the strict comparison after cross multiplication. This same example exceeds the pure linear-rotation envelope $E_{10}(1/6) \approx 0.00774841$. It therefore rules out an elementary repair considered in this note: taking only the maximum of N^{-2} and the rotation envelope. This repair is not attributed to an external conjecture. No complete mixed-class envelope is claimed.

The construction combines a rotation with a constant hinge subgradient over part of the trajectory. An exact terminal active-set analysis and numerical performance-estimation work guided its choice. Neither the discovery computation nor the exploratory optimisation is needed to replay the finite certificate.

Appendix A gives an analytic family that already violates the unit-relaxation bound at $N = 9$, with a fully rational instance at parameter $t = 1/100$. It explains how a small rotation perturbs a scalar hinge extremiser. The small exact gap exposes an active-set mechanism that approximate arithmetic could miss; it is not evidence of practical instability, divergence or failure of optimisation. No minimal failure horizon is asserted.

7. Literature boundary and remaining questions

What is established here

Propositions 1 and 2 refute the exact conjecture as printed. Theorem 4 establishes the correct uniform exponent for precisely that operator class and all fixed admissible parameters. Theorem 5 describes the whole linear subclass under prescribed scalar schedules. Theorem 6 gives an exact minimax synthesis for the nonlinear $A = 0$ subclass under arbitrary real adaptive scalar relaxation. Corollary 6.1 identifies the linear-subclass infimum for conventional-range adaptive scalars as an elementary consequence. The hinge certificate excludes a natural but incomplete finite-horizon repair considered in this note.

What is inherited from earlier work

The resolvent/cocoercivity correspondence is standard, with directly relevant formulations in Giselsson (2017). The sharp firmly nonexpansive residual constant comes from Gu and Yang (2019) and related work of Lieder. Von Neumann's polynomial inequality is classical. Contreras and Cominetti (2022) establish an order barrier for prescribed Krasnoselskii–Mann schedules in general normed spaces. Sarkar and Aggarwal (2026) explicitly supply the planar rotation

product used in a recent fixed-schedule lower bound. None of these ingredients is presented as a new discovery.

The mathematical statement of Theorem 6 is more specific than these inspected statements: its normalisation uses initial distance in Euclidean space, its constant is exact, and its lower bound holds on one horizon-dependent instance for every real adaptive policy. Its derivation is nevertheless short once those earlier ingredients are combined. This limits the originality claim.

Component	Contribution and inherited boundary
Propositions 1–2	Explicit refutations of the printed conjecture; the rotational mechanism itself is established mathematics
Theorem 4	Class-specific sharp exponent from a tailored admissible lower family and the standard averaged-map upper estimate
Theorem 5	Linear-envelope corollary of the classical contraction polynomial inequality
Theorem 6	Adaptive-policy formulation and penalty identity synthesized with an inherited sharp proximal-point upper bound; historical overlap remains bounded by source access
Corollary 6.1	Elementary restricted-policy linear consequence, not a new general nonlinear rate
Hinge witnesses	Exact finite mixed-class obstruction and analytic perturbation mechanism; no exact mixed-class envelope or minimal failure horizon

Search scope and missingness

The initial search date was 1 November 2026, with the target version, proximal-point source, schedule literature and named access gap rechecked on 5 November 2026. The search covered the originating arXiv versions and reference list, the publisher and Optimization Online records, two general web-search systems, exact conjecture-number and title searches, formula searches for the rotation and residual products, and the aliases cocoercive equation, averaged operator, Krasnoselskii–Mann iteration, relaxed proximal point, optimal relaxation, and step-schedule acceleration. It also checked the current Evidence Press index for duplicate work. The complete query families, inspected-source notes and claim decisions accompany this paper.

No prior refutation of Conjecture 3.5 was located. This does not prove that no such refutation exists. He et al. (2021), on optimal relaxation parameters, was available only through its abstract and bibliographic record. Its exact overlap with the adaptive minimax formulation remains unverified. Refreshed lawful publisher and repository retrieval attempts did not obtain the full text. We therefore make no historical-priority claim for Theorem 6 or its elementary restricted-policy corollary; their role here is a precisely scoped synthesis accompanying the conjecture refutation. No author enquiry, subscription-database search or unaffiliated specialist review occurred. The negative search evidence supports bounded priority language only.

Questions left open

We do not resolve Conjectures 5.1, 5.6 or 5.7 of the originating paper. Those concern objective values for convex composite optimisation, including silver schedules and inertial acceleration. The

rotational counterexample in Proposition 2 does not satisfy their gradient hypothesis and cannot refute them. The scalar example in Proposition 1 is also not a refutation of those objective bounds.

Additional numerical performance-estimation checks reproduced the proposed objective bounds at the finite horizons recorded in the exploratory notes. Such agreement is not a proof. A mirror-coordinate reformulation exposes the difficulty of a general inertial proof, but does not close it.

We also leave unresolved the exact mixed-class finite-horizon supremum, the full nonlinear minimax under the restriction $0 < \lambda_j < 2$, and any fixed-instance asymptotic strengthening. These are different questions from the proved uniform exponent and the enlarged-policy minimax theorem.

8. Verification and reproducibility

The main refutations use rational input and can be checked without a numerical optimiser. The bundle supplies two separately authored exact implementations, a claim/check ledger, an informal proof outline, a rating-free assessment and revision response, source notes and a manifest. The primary certificate uses Python's standard-library `fractions.Fraction`; rational equality and inequality checks do not depend on floating-point tolerances. It reconstructs the updates from the original resolvent and proximal definitions.

The general theorems are written mathematical arguments. Finite checks support their algebra and indexing but do not establish their universal quantifiers. No Lean, Rocq or Isabelle proof was produced. An independent AI agent re-derived the central rational matrix identities and the schedule penalty; this is internal cross-checking, not journal peer review or unaffiliated mathematical validation.

To replay the exact checks from the extracted package, run:

```
PYTHONDONTWRITEBYTECODE=1 python3 -B run_checks.py
PYTHONDONTWRITEBYTECODE=1 python3 -B -O run_checks.py
```

The replay record must identify the exact script versions and whether normal and optimized execution completed or were rejected. Separately authored implementations within this project remain producer-side implementation diversity, not unaffiliated reproduction. The analytic Appendix A argument, its exact $t = 1/100$ certificate and the optional symbolic consistency check have different scopes.

A basic sanity check separates the error measures. The conjecture concerns a **squared** residual: its proposed N^{-2} corresponds to residual norm N^{-1}. Theorem 4's sharp squared-residual order N^{-1} corresponds to residual norm $N^{-1/2}$. The counterexample does not claim failure of convergence.

Availability and declarations

The accompanying evidence package contains the source manuscript, exact-check scripts, retained outputs and source-comparison notes. It uses constructed mathematical examples, not personal or human-participant data; human-subject ethical approval is not applicable. Mathematical development, exposition and internal checking were AI-assisted. The scholarly byline is Anonymous; publication authority does not imply a human mathematical contribution. No additional human CRediT role is asserted. No external research grant or competing interest has been reported in the supplied provenance; these are reporting limits, not independently verified absence claims.

Appendix A. An analytic nine-evaluation hinge family

Proposition A.1. At $\beta = 1$, $\gamma = 1/2$ and $\lambda = 1$, there is a rational-parameter family of two-dimensional admissible instances for which

$$\frac{\|T(w_t^9) - w_t^9\|^2}{\|w_t^1\|^2} = \frac{1}{81} + \frac{5}{729}t^2 + O(t^4) \quad \text{as } t \downarrow 0. \tag{A.1}$$

Consequently the proposed bound fails for every sufficiently small positive t in this family.

Construction. Write $e = (1,0)^\mathsf{T}$ and $H = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Define

$$B_t = \frac{t^2 I + tH}{1+t^2}, \qquad Q_t = (I + B_t/2)^{-1} = \frac{1}{4+9t^2}\begin{pmatrix} 4+6t^2 & 2t \\ -2t & 4+6t^2 \end{pmatrix},$$

$$\begin{aligned} R_t &= 2Q_t - I, & S_t &= \sum_{j=0}^{7} Q_t^j, \\ D_t &= e^\mathsf{T} R_t S_t e + 1, & w_t^1 &= (1, t/2)^\mathsf{T}, \\ \delta_t &= \frac{e^\mathsf{T} R_t Q_t^8 w_t^1}{D_t}, & g_t(x) &= 2\delta_t \max\{x_1, 0\}. \end{aligned} \tag{A.2}$$

Every expression is rational for rational t. Moreover, $B_t^\mathsf{T} B_t = (B_t + B_t^\mathsf{T})/2$, so B_t is 1-cocoercive. For $t > 0$ it is strongly monotone. Once $\delta_t > 0$, the function g_t is convex and the origin is the unique solution and fixed point.

Proximal branches. All quantities in (A.2) are analytic near zero because $D_0 = 9$. In particular, $\delta_0 = 1/9 > 0$. Consider the candidate affine updates

$$w^{k+1} = Q_t w^k - \delta_t e.$$

At $t = 0$, their proximal outputs have first coordinates $1 - k/9$, which are strictly positive for $k = 1, \dots, 8$. Continuity preserves those eight active branches for all sufficiently small positive t. The definition of δ_t gives the exact terminal equality

$$e^\mathsf{T}\left[R_t(Q_t^8 w_t^1 - \delta_t S_t e) - \delta_t e\right] = 0.$$

Thus the ninth proximal input has first coordinate $\delta_t > 0$, exactly the endpoint of the zero branch in (18), and its proximal output has first coordinate zero. Induction proves that the candidate affine updates coincide with the actual DRS iteration through the ninth update.

Residual calculation. The final residual norm is $\|K_t w_t^1\|$, where

$$K_t = Q_t^8\left[I - Q_t + \frac{e(e^\mathsf{T} R_t Q_t^8)}{D_t}\right]. \tag{A.3}$$

Indeed, $w_t^9 = Q_t^8 w_t^1 - \delta_t S_t e$ and $(I - Q_t)S_t = I - Q_t^8$. Substituting these identities into $(I - Q_t)w_t^9 + \delta_t e$ proves (A.3).

Expanding the rational matrices gives

$$Q_t^8 = I - 4tH - 13t^2 I + O(t^3), \qquad S_t = 8I - 14tH - 35t^2 I + O(t^3),$$

$$D_t = 9 - 61t^2 + O(t^4),$$

and hence

$$K_t = \begin{pmatrix} 1/9 + t^2/324 & t/18 \\ t/18 & 19t^2/36 \end{pmatrix} + \begin{pmatrix} O(t^4) & O(t^3) \\ O(t^3) & O(t^4) \end{pmatrix}.$$

Multiplying by $w_t^1 = (1, t/2)^\mathsf{T}$, squaring and dividing by $1 + t^2/4$ gives (A.1). The even remainder follows from the parity of the rational matrices: diagonal entries are even and off-diagonal entries are odd in t. Since $5/729 > 0$, the strict violation follows. □

The standard-library certificate **optimisation_hinge_n9_exact.py** checks $t = 1/100$, including the eight strict branch inequalities, the ninth exact boundary equality and the strict comparison to $1/81$. Its ratio to the conjectured bound is approximately 1.00005518699093. The expansion also has a separate symbolic consistency check. Neither result establishes a minimal possible failure horizon or an exact mixed-class envelope.

References

Target and operator-splitting foundations

Abbaszadehpeivasti, H., & Zamani, M. (2025). *On the convergence rate of the Douglas–Rachford splitting algorithm* (Version 2) [Preprint]. arXiv. https://arxiv.org/abs/2509.06676v2

Giselsson, P. (2017). Tight global linear convergence rate bounds for Douglas–Rachford splitting. *Journal of Fixed Point Theory and Applications, 19*, 2241–2270. https://doi.org/10.1007/s11784-017-0417-1

Gu, G., & Yang, J. (2019). *Optimal nonergodic sublinear convergence rate of proximal point algorithm for maximal monotone inclusion problems* (Version 2) [Preprint]. arXiv. https://arxiv.org/abs/1904.05495v2

Lieder, F. (2021). On the convergence rate of the Halpern-iteration. *Optimization Letters, 15*, 405–418. https://doi.org/10.1007/s11590-020-01617-9

Scheduling and nearby prior art

Contreras, J. P., & Cominetti, R. (2022). *Optimal error bounds for nonexpansive fixed-point iterations in normed spaces* (Version 3) [Author manuscript]. https://arxiv.org/abs/2108.10969v3

He, S., Dong, Q.-L., Tian, H., & Li, X.-H. (2021). On the optimal relaxation parameters of Krasnosel'ski–Mann iteration. *Optimization, 70*(9), 1959–1986. https://doi.org/10.1080/02331934.2020.1767101 [Abstract inspected; full text unavailable.]

Sarkar, D., & Aggarwal, V. (2026). *Non-expansive two-time-scale stochastic approximation: A fixed-schedule one-quarter barrier and bias-corrected acceleration* [Preprint]. arXiv. https://arxiv.org/abs/2607.13414

Classical polynomial bound

Hartz, M. (2025). On von Neumann's inequality on the polydisc. *Mathematische Annalen, 391*, 5235–5264. https://doi.org/10.1007/s00208-024-03040-2

von Neumann, J. (1950). Eine Spektraltheorie für allgemeine Operatoren eines unitären Raumes. *Mathematische Nachrichten, 4*(1–6), 258–281. https://doi.org/10.1002/mana.3210040124 [Year follows the publisher metadata; the work is also conventionally cited as 1951. The contraction inequality was inspected in Hartz (2025).]